

The Role of Alloyed Molybdenum in Austenitic Stainless Steels in the Inhibition of Pitting in Neutral Halide Solutions*

KATSUHISA SUGIMOTO AND YOSHINOBU SAWADA*

Abstract

In the passive region of austenitic stainless steels alloyed with Mo, the formation of MoO_4^{2-} ions can be expected in neutral halide solutions by the transpassive dissolution of Mo. It has been shown that MoO_4^{2-} ions added to neutral NaCl solutions act as an effective inhibitor against pitting of austenitic stainless steels with and without Mo. The interaction between alloyed Mo in the steels and added MoO_4^{2-} ions in the solutions is appreciable. It is likely that the inhibition of pit growth by the adsorption of MoO_4^{2-} ions which are thought to result from the dissolution of the steels at the initial stage of pitting leads to increased pitting resistance of austenitic stainless steels containing Mo.

It is a well known fact that the addition of Mo to austenitic stainless steels increases pitting resistance because of shifts of the pitting potential in the noble direction.¹⁻⁶ Laboratory experiments, however, show that the factors which lead to the increase in the pitting resistance of the steel by alloying Mo are very complex. For instance, the beneficial effect of Mo addition decreases with increase in temperature,⁷ and the critical temperature at which pitting initiates depends on the Mo content of the steel and the Cl⁻ concentration of the solution used.⁸ It has been reported that alloyed Mo is ineffective to inhibit bromide pitting.^{4,9,10} Furthermore, published data indicating some differences between the effects of Mo additions on austenitic stainless steel and ferritic stainless steel have been presented.¹¹

Although the action of Mo in regard to improving the resistance of austenitic stainless steels against pitting is not known precisely, we have proposed the following 2 mechanisms to account for the inhibition: (1) the inhibition of pit initiation by an improvement of the properties of the passive film for protection against invasion of Cl⁻ ions, and (2) the inhibition of pit growth by the adsorption of MoO_4^{2-} ions, which are thought to be concentrated within a pit, by the dissolution of the alloyed Mo at the initial stage of pitting.

Our recent X-ray photoelectron spectroscopy (ESCA) studies on the passive film compositions of austenitic stainless steels containing Mo have shown that Mo is one of the essential constituents of the passive film, and that the Mo content of the film increases with increase in Mo content of the steels.¹² Therefore, it would be reasonable to expect that the contribution of Mo in the passive film is the cause of the improved pitting resistance of steel. It is uncertain, however, whether the increase in pitting resistance is brought about only by modifications of the properties of the passive film, because the alloyed Mo may dissolve as MoO_4^{2-} ions by the transpassive reaction of Mo in the passive region of the steel. Furthermore, it is difficult to explain by the film theory why the alloying addition of Mo is ineffective in a NaBr solution.

Although knowledge of the contribution of MoO_4^{2-} ions to the mechanism of pitting inhibition for the Mo-containing austenitic stainless steel is insufficient, it has previously been reported that MoO_4^{2-} ions added to a neutral NaCl solution effectively inhibit the pitting dissolution of an austenitic stainless steel that does not contain Mo.¹³ Suzuki, *et al.*, suggested the possibility of a higher concentration of MoO_4^{2-} ions in an active pit than in the bulk solution for Type 316L and an 18Cr-16Ni-5Mo steel.¹⁴

The purpose of the present study is to clarify the common feature in both methods for pitting inhibition, that is, the alloying

TABLE 1 — Composition of the Steels Studied (Wt%)

Specimen	C	Si	Mn	P	S	Cr	Ni	Mo
20Cr-25Ni	0.053	0.54	1.53	0.009	0.010	20.06	25.58	0.01
20Cr-25Ni-1Mo	0.057	0.48	1.48	0.009	0.010	19.99	25.88	1.05
20Cr-25Ni-3Mo	0.064	0.52	1.47	0.009	0.010	20.16	25.52	2.97
20Cr-25Ni-5Mo	0.059	0.46	1.44	0.008	0.010	19.96	25.66	5.06

addition of Mo to steels and the addition of MoO_4^{2-} ions to halide solutions, and to speculate on the role of alloyed Mo in the inhibition of pitting. To achieve this, anodic polarization curves and ESCA spectra were measured using several combination of austenitic stainless steels with and without Mo, and halide solutions with and without MoO_4^{2-} ions.

Experimental

Rolled sheets of austenitic 20Cr-25Ni steels containing 0 to 5 Wt% Mo and pure Mo (99.99 Wt%) were used as specimens. Analytical compositions of the stainless steels used are listed in Table 1. Specimens $40 \times 20 \times 1.5$ (mm) were used for electrochemical polarization measurements, and $18 \times 5 \times 0.3$ (mm) for ESCA measurements. All of these specimens were annealed at 1050°C for 2 hours in vacuo and then water-quenched at 0°C.

Specimen surfaces were mechanically polished with emery papers up to No. 6/0 and with cloth using the diluted suspension of chromium oxide, degreased with absolute ethyl alcohol, washed in distilled water, and finally dried with air blasts. Specimens for the electrochemical polarization measurements were masked with Teflon seal tapes attached to an epoxy resin undercoating, except for an area 1 cm^2 while those for ESCA measurements were not masked.

As test solutions, 0.05 to 0.1M-NaCl, 0.1M-NaBr, NaCl + Na_2MoO_4 (NaCl : 0.05 ~ 0.5M, Na_2MoO_4 : 0.02 ~ 0.2M) and NaBr + Na_2MoO_4 (NaBr : 0.1M, Na_2MoO_4 : 0.2 ~ 1.0M) were used. All the solutions used were approximately neutral; for example, the pH values of 0.1M-NaCl, 0.1M-NaCl + 0.1M- Na_2MoO_4 , and 0.1M-NaCl + 0.2M- Na_2MoO_4 were 6.16, 6.68, and 7.27, respectively. The solutions were deaerated by bubbling pure nitrogen for more than 30 minutes.

In the measurement of anodic polarization curves, the electrode potential was regulated by a potentiostat and current recording was made after holding the potential for 1 minute at the setting of the potential which was increased successively from the corrosion potential in steps of 25 mV. In this paper, the potentials of test electrodes are represented with respect to the standard hydrogen electrode (NHE), although the values were measured using a saturated calomel reference electrode. Measurements were made at

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*Department of Metallurgy, Faculty of Engineering, Tohoku University, Sendai, Japan

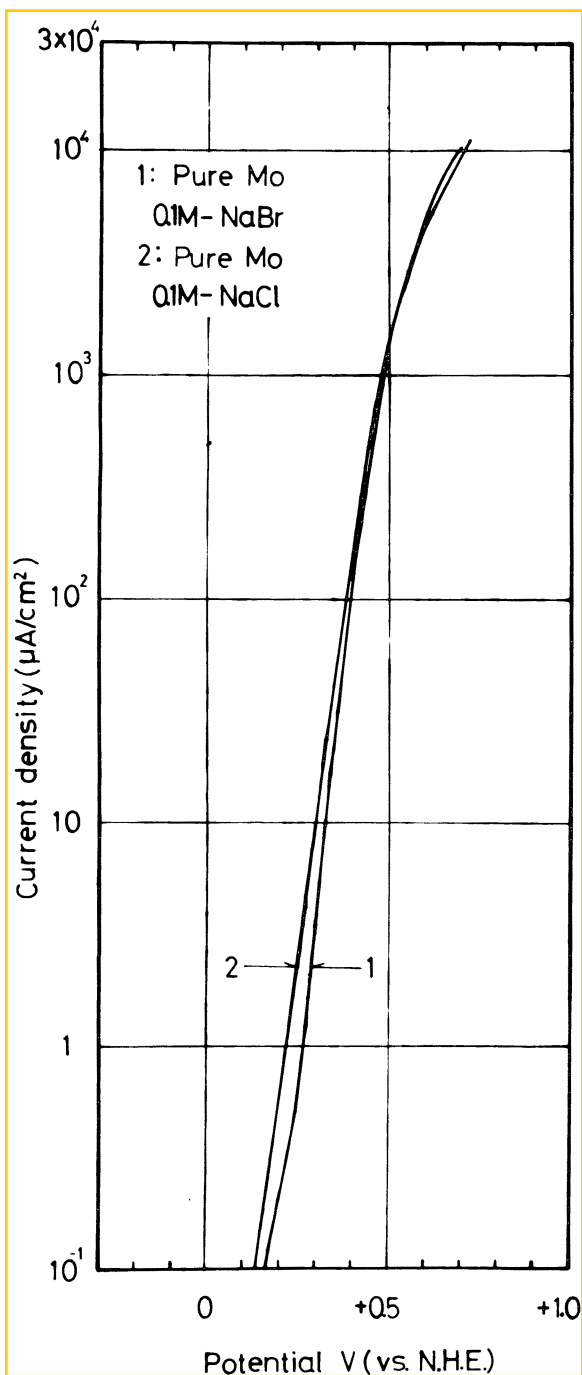


FIGURE 1 – Anodic polarization curves for pure Mo in 0.1M-NaCl and 0.1M-NaBr at 25 C.

various temperatures in the range 10 to 70 C under a stationary-state condition

Surface analyses of specimens which were treated anodically in chloride solutions were performed by means of ESCA using a Kokusai Denki type KES-X-2001 electron spectrometer. The aluminum $K_{\alpha 1,2}$ X-ray generated at 1486.6 eV was used for excitation. The voltage and the current applied to the X-ray tube were 10 kV and 20 mA, respectively. The specimen container was evacuated to 10^{-7} Torr and held at room temperature. The photoelectron spectra of the $2p_{3/2}$ and $2p_{1/2}$ levels for Fe, Cr, Ni and Cl, the $3d_{3/2}$ and $3d_{5/2}$ levels for Mo, and the 1s level for O and C were measured. The binding energies at the peaks of the photoelectron spectra obtained were corrected by the energy of C 1s of the background spectrum as a reference, assuming the energy difference between the 1s level for C and the Fermi level for metallic Pd to be 285.0 eV. The valencies of elements in passive

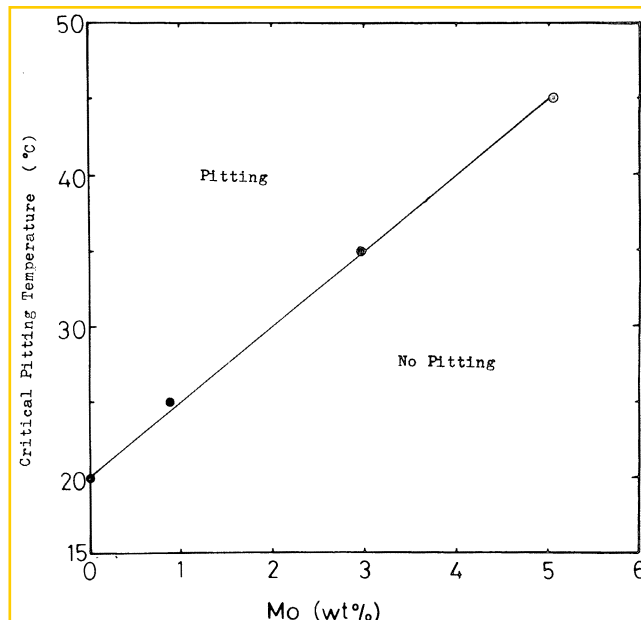


FIGURE 2 – Critical pitting temperature as a function of the Mo content of 20Cr-25Ni steels in 0.1M-NaCl

films were evaluated by comparing the binding energies at the peaks of the photoelectron spectra of passive films and several kinds of pure compounds¹⁵

Results

Anodic Dissolution of Pure Mo in NaCl and NaBr Solution

Figure 1 shows anodic polarization curves for pure Mo in 0.1M-NaCl and 0.1M-NaBr. Corrosion potentials of pure Mo in these solutions were nearly equal, that is, approximately +0.15 V. The general dissolution occurred in both solutions with a steep increase in current as the electrode potential was changed from the corrosion potential to higher values. The general dissolution in this case is probably caused, not by the pitting dissolution, but by the transpassivation reaction, since pure Mo showed the same mode of dissolution when anodically polarized in 1M-Na₂SO₄ from a corrosion potential of approximately +0.15 V.

Effect of Temperature on Pitting of Mo-Containing Steels in NaCl Solution

The effect of temperature on anodic polarization curves of 20Cr-25Ni steels containing 1 to 5 Wt% Mo in 0.1M-NaCl was determined.

Figure 2 shows the relation between the critical pitting temperature and the Mo content of the steels. Below these critical temperatures for the pitting initiation, each steel maintained passivity to about +1.15 V until the transpassive dissolution occurred with the accompanying oxygen evolution. The critical pitting temperature (T_{pit}) became higher with increase in Mo content. This is defined as

$$T_{pit} = 5 [\text{Mo}\%] + 20 \quad (^\circ\text{C}). \quad (1)$$

The critical temperature depended on NaCl concentration too and were observed to decrease with increasing concentration.

Effect of Temperature on Pitting of Mo-free Steels in NaCl + Na₂MoO₄ Solutions

It has previously been reported that MoO_4^{2-} ions added to a NaCl solution act as an effective inhibitor against pitting of austenitic stainless steels.¹³ In the present work, the authors investigated the temperature range in which MoO_4^{2-} ions show an

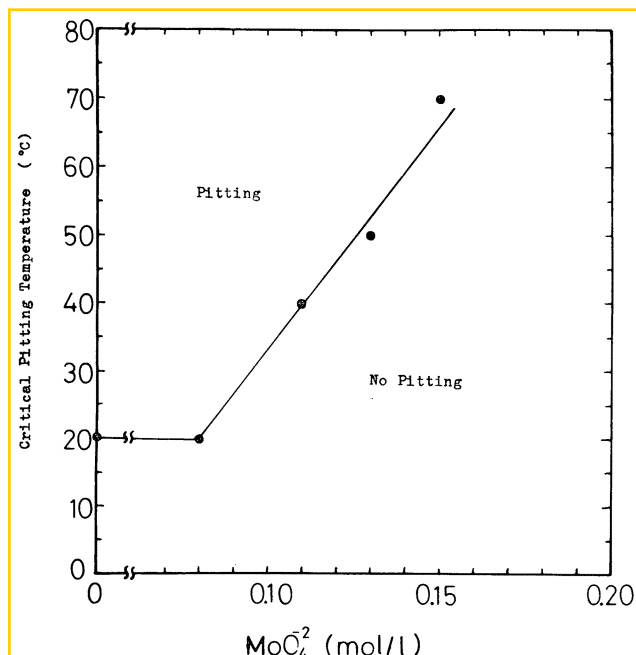


FIGURE 3 – Critical pitting temperature of 20Cr-25Ni steel as a function of the MoO_4^{2-} concentration in 0.1M-NaCl

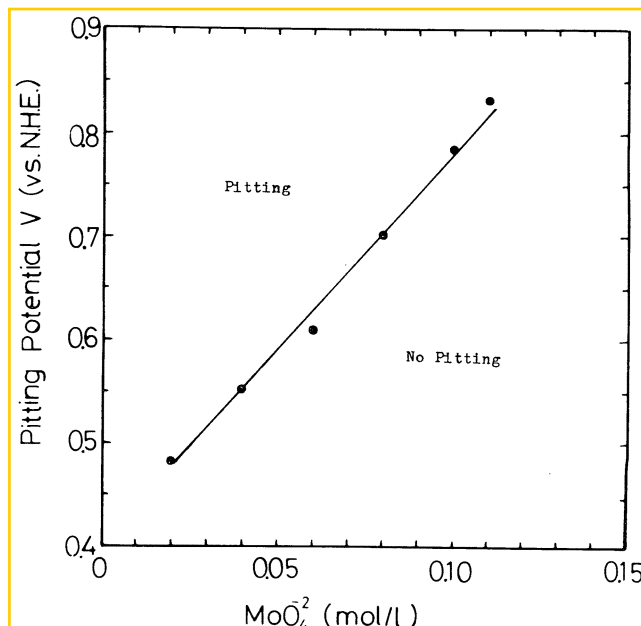


FIGURE 4 – Pitting potential of 20Cr-25Ni steel as a function of the MoO_4^{2-} concentration in 0.1M-NaCl at 50 C.

effective inhibition, the change in pitting potential with the MoO_4^{2-} concentration, and the relation between the chloride concentration and the critical concentration of MoO_4^{2-} ions required to completely inhibit pitting

Figure 3 shows the effect of MoO_4^{2-} concentration on the critical pitting temperature of the 20Cr-25Ni steel in 0.1M-NaCl. The critical temperature (T_{pit}) increased linearly with the MoO_4^{2-} concentration. The linear relation is expressed as

$$T_{\text{pit}} = 657.1 (\text{MoO}_4^{2-}, \text{mol/L}) - 32.28 \quad (^\circ\text{C}). \quad (2)$$

Pitting on the steel in 0.1M-NaCl below 70 C was completely inhibited when the solution contained MoO_4^{2-} ions more than 0.2M. When the complete inhibition is achieved, the 20Cr-25Ni steel retains passivity to about +1.15 V, and then the transpassive dissolution occurs, accompanying oxygen evolution

Figure 4 shows the change in the pitting potential of the 20Cr-25Ni steel with the MoO_4^{2-} concentration in 0.1M-NaCl at 50 C. The pitting potential was shifted linearly in the noble direction with increase in MoO_4^{2-} concentration up to 0.12M, over which pitting was completely inhibited at 50 C.

Figure 5 shows the relation between the Cl^- concentration and the minimum concentration of MoO_4^{2-} ions which is required to inhibit pitting of the 20Cr-25Ni steel at 25 C. The linear relation given in Equation (3) is obtained in the log-log plot between the Cl^- and the MoO_4^{2-} concentration (mol/l).

$$\log(\text{Cl}^-) = 1.17 \log(\text{MoO}_4^{2-}) + 0.079. \quad (3)$$

Interaction in Pitting Inhibition Between Mo in Steel and MoO_4^{2-} Ions in Solutions

Anodic polarization behavior of Mo-containing austenitic steels in NaCl solutions with MoO_4^{2-} ions shows that Mo in the steels and MoO_4^{2-} ions in the solutions interact with each other to inhibit pitting. Figure 6 shows the anodic polarization curves for the 20Cr-25Ni-3Mo steel in a 0.1M-NaCl + 0.11M- Na_2MoO_4 solution at 50 C and the 20Cr-25Ni-5Mo steel in the same solution at 60 C. Both curves show stable passivity up to the transpassive potential. As a matter of course, pitting was not observed in either case. As previously stated, both steels used here underwent pitting in 0.1M-NaCl without MoO_4^{2-} ions at 50 or 60 C (Figure 2), and the addition of MoO_4^{2-} ions up to 0.11M to 0.1M-NaCl above 50 C was

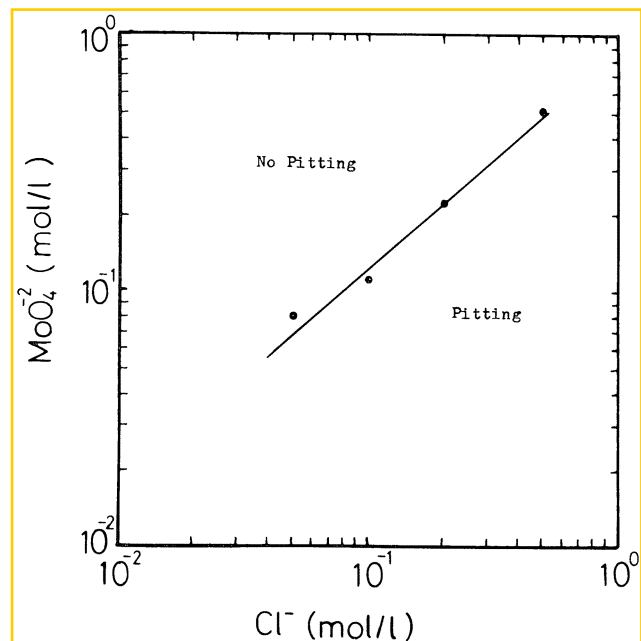


FIGURE 5 – Effect of the Cl^- concentration on the minimum concentration of MoO_4^{2-} ions which is necessary for pitting inhibition of 20Cr-25Ni steel at 25 C.

insufficient to inhibit pitting of the 20Cr-25Ni steel without Mo (Figure 3). As shown in Figure 6, however, a simultaneous application of Mo alloying and MoO_4^{2-} addition inhibited pitting effectively, as against a separate use of these procedures by which pitting cannot be eliminated. These results suggest that there exists a common feature in the pitting inhibition mechanism between alloyed Mo and MoO_4^{2-} ions added to the solution.

Pitting Behavior of 20Cr-25Ni and 20Cr-25Ni-5Mo Steels in NaBr Solution

Anodic polarization curves for 20Cr-25Ni and 20Cr-25Ni-5Mo steels in 0.1M and 1.0M-NaBr at 25 C are shown in Figure 7. The effect of Mo content on the pitting potential was scarcely observed

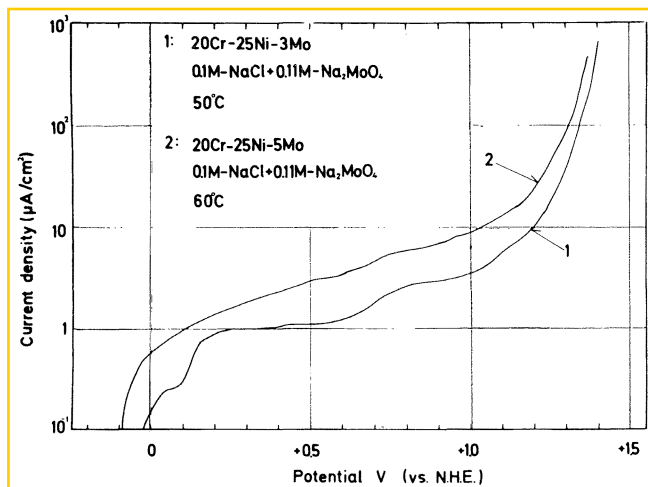


FIGURE 6 — Anodic polarization curves for 20Cr-25Ni-3Mo and 20Cr-25Ni-5Mo steels in a 0.1M-NaCl + 0.11M-Na₂MoO₄ solution at 50 and 60°C.

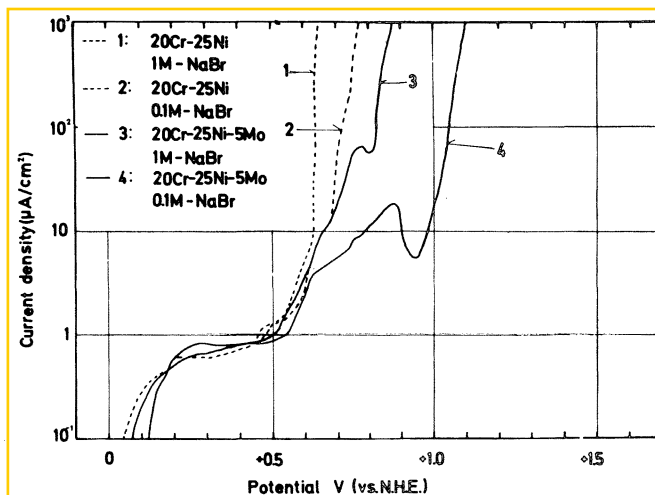


FIGURE 7 — Anodic polarization curves for 20Cr-25Ni and 20Cr-25Ni-5Mo steels in 0.1M and 1M-NaBr at 25°C

since the pitting potentials of both steels were almost equal in the solutions of the same NaBr concentration. For example, the pitting potential for both steels was +0.521 V in 0.1M-NaBr. The effect of alloyed Mo, however, was clearly observed on the rate of pitting dissolution in the solutions. That is, an increase in the Mo content decreases the rate, since the anodic polarization curve of Figure 7 for the 20Cr-25Ni-5Mo steel in 0.1M-NaBr represents a larger polarization than that for the 20Cr-25Ni steel in the same solution and a temporary decrease in current in the potential range higher than the pitting potential.

Then, the inhibiting efficiency of MoO₄²⁻ ions for the bromide pitting of the 20Cr-25Ni steel without Mo was examined in 0.1M-NaBr containing MoO₄²⁻ ions up to 1.0M at 25°C. Increasing the MoO₄²⁻ concentration in the solution shifted the pitting potential of the steel in the noble direction and made the polarization of the anodic curve larger. This means that MoO₄²⁻ ions act as an inhibitor against pitting of the steel in NaBr solutions. However, the inhibiting efficiency of MoO₄²⁻ ions for pitting in NaBr solutions was extremely low compared with the case of NaCl solutions. For example, at a temperature of 25°C, it was impossible to prevent the initiation of pitting in 0.1M-NaBr above +0.97 V even if the solution contained 1.0M-MoO₄²⁻; on the other hand, the complete inhibition was achieved in 0.1M-NaCl by adding only 0.11M-MoO₄²⁻.

X-ray Photoelectron Spectra of Passive Films

The resistance of austenitic stainless steels against pitting in NaCl solutions is expected to depend on the chemical properties of their passive films formed in the solutions. The differences in the chemical composition of surface films between the specimens susceptible or not susceptible to pitting in NaCl solutions were

TABLE 2 — Passivation Treatments for Specimens Used for X-ray Photoelectron Spectroscopic Analyses

Symbol of Specimen	Material	Passivation Treatment
A	20Cr-25Ni	0.1M-NaCl (20°C), +0.946 V, 60 min
B	20Cr-25Ni-5Mo	0.1M-NaCl (20°C), +0.946 V, 60 min
C	20Cr-25Ni	0.1M-NaCl + 0.2M-Na ₂ MoO ₄ (20°C), +0.946 V, 60 min
D	20Cr-25Ni-5Mo	Passivated in the air at room temperature
Mo sheet	99.99% Mo	Freshly polished with emery paper

examined on the basis of the photoelectron spectra. The stainless steel specimens used for the measurement of photoelectron spectra are given in Table 2. Specimen A was a Mo-free 20Cr-25Ni steel on which pitting occurred in a NaCl solution. Specimen B was the 20Cr-25Ni-5Mo steel which was resistant to pitting in the solution. Specimen C was a Mo-free 20Cr-25Ni steel which was prevented from pitting by adding MoO₄²⁻ ions to the solution. Specimen D was the 20Cr-25Ni-5Mo steel with air-formed films which was used for comparison. The values of binding energy at the peaks of the photoelectron spectra for the specimens listed in Table 2 and reference pure compounds are tabulated in Table 3.

Discussion

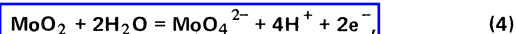
Anodic Polarization Behavior of Pure Mo in Neutral Halide Solutions

It had been said that the passive film of pure Mo was composed of MoO₂ and the reaction product of the transpassivation was MoO₄²⁻ ions.¹⁶⁻¹⁸ Accordingly, the transpassivation reaction for Mo in a neutral solution with pH higher than 6, and its equilibrium potential, may be represented by the following equations, respectively.¹⁹

TABLE 3 — Binding Energies for the Peaks Observed in the X-ray Photoelectron Spectra

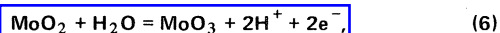
Specimen	Fe 2p _{3/2}		Cr 2p _{3/2}		Ni 2p _{3/2}		Mo 3d _{3/2} and 3d _{5/2}				Cl 2p _{1/2} and 2p _{3/2}	
	Oxide		Oxide		Oxide		Oxide				Metal	
	3d _{3/2}	3d _{5/2}	3d _{3/2}	3d _{5/2}	3d _{3/2}	3d _{5/2}	3d _{3/2}	3d _{5/2}	3d _{3/2}	3d _{5/2}	Q 1s	2p _{1/2} 2p _{3/2}
A	712.9 ⁽¹⁾	708.0 ⁽²⁾	577.4 ⁽¹⁾	574.1 ⁽³⁾	861.1 ⁽²⁾	855.5 ⁽¹⁾	235.5 ⁽²⁾	232.2 ⁽¹⁾	230.9 ⁽³⁾	227.8 ⁽²⁾	532.1 ⁽³⁾ 530.5 ⁽¹⁾	199.8 ⁽³⁾ 198.2 ⁽¹⁾
B	712.1 ⁽¹⁾	707.7 ⁽²⁾	577.1 ⁽¹⁾	574.1 ⁽³⁾	861.6 ⁽²⁾	856.3 ⁽¹⁾	235.6 ⁽³⁾	232.3 ⁽¹⁾	230.7 ⁽¹⁾	227.7 ⁽²⁾	532.0 ⁽³⁾ 530.5 ⁽¹⁾	202.0 ⁽³⁾ 200.2 ⁽¹⁾
C	712.2 ⁽¹⁾	707.8 ⁽²⁾	577.6 ⁽¹⁾	574.8 ⁽³⁾	862.6 ⁽²⁾	856.2 ⁽¹⁾	235.6 ⁽³⁾	232.3 ⁽¹⁾	230.7 ⁽¹⁾	227.7 ⁽²⁾	532.5 ⁽³⁾ 530.8 ⁽¹⁾	201.8 ⁽³⁾ 200.8 ⁽¹⁾
D	712.2 ⁽¹⁾	707.4 ⁽²⁾	577.5 ⁽¹⁾	574.2 ⁽³⁾	857.4 ⁽²⁾	853.6 ⁽¹⁾	235.2 ⁽²⁾	232.2 ⁽¹⁾	230.7 ⁽¹⁾	227.7 ⁽²⁾	531.8 ⁽¹⁾ 530.0 ⁽³⁾	
Mo sheet												
MoO ₃							235.8 ⁽²⁾	232.7 ⁽¹⁾	231.0 ⁽²⁾	227.9 ⁽¹⁾	533.3 ⁽³⁾	531.3 ⁽¹⁾
Na ₂ MoO ₄ · 2H ₂ O							235.3 ⁽²⁾	232.2 ⁽¹⁾				531.1 ⁽¹⁾
NaCl												530.7 ⁽¹⁾
												200.0 ⁽³⁾ 198.5 ⁽¹⁾

(1) = Strong
(2) = Weak
(3) = Should



$$E = 0.606 - 0.1182\text{pH} + 0.0295 \log a_{\text{MoO}_4^{2-}} \quad (5)$$

As the pH of 0.1M-NaCl used is 6.16, MoO_4^{2-} ions should be more stable than MoO_2 in the potential region above -0.199 V, assuming $a_{\text{MoO}_4^{2-}} = 10^{-6}$. It is possible to assume that the oxide film of MoO_3 is formed on the Mo electrode in the potential region of transpassivity according to Equation (6):



$$E = 0.320 - 0.0591 \text{pH} \quad (7)$$

The MoO_3 film, however, even if formed in this solution, will be porous and not depress the anodic dissolution of the electrode because the oxide MoO_3 dissolves as HMoO_4^- or MoO_4^{2-} ions in solutions with pH higher than 3.7.¹⁹

The general dissolution of pure Mo observed in 0.1M-NaCl and 0.1M-NaBr when polarized in the noble direction from the corrosion potential (about +0.15 V) is thought to be the transpassive dissolution caused by the reaction of Equation (4). The passive region of the 20Cr-25Ni steels containing 1 to 5 Wt% Mo in 0.1M-NaCl corresponds to the transpassive region of pure Mo (because the corrosion potentials of these steels in the solution are about +0.1 V). MoO_4^{2-} ions, therefore, should be formed if these steels are dissolved by pitting in their passive regions.

Pitting Inhibition Mechanism of MoO_4^{2-} Ions Added to Neutral NaCl Solutions

On addition of MoO_4^{2-} ions to NaCl solutions, the following 2 pitting inhibition mechanisms may be considered for austenitic stainless steels without Mo: (1) MoO_4^{2-} ions are adsorbed preferentially on the steel surface and preclude the adsorption of Cl^- ions and (2) the MoO_3 film is formed on the steel surface by the reduction of MoO_4^{2-} ions, and this film prevents the chloride attack. Since the pH values of $\text{NaCl} + \text{Na}_2\text{MoO}_4$ solutions used are approximately 7, the equilibrium potential for the reaction represented by Equation (4) is -0.251 V as calculated from Equation (5). Consequently, the reduction of MoO_4^{2-} ions to MoO_2 is improbable in the potential region of passivity for the 20Cr-25Ni steel, because corrosion potentials of the steel in $\text{NaCl} + \text{Na}_2\text{MoO}_4$ solutions which are approximately +0.05 V, are higher than the equilibrium potential for Equation (4). For the 20Cr-25Ni steel passivated in a 0.1M-NaCl + 0.2M- Na_2MoO_4 solution, the X-ray photoelectron spectra studies showed that Mo is oxidized to about Mo(VI) in the passive film, as mentioned later. Consequently, the mechanism which assumes the formation of the Mo(IV) oxide, MoO_2 , cannot be considered valid. On the other hand, the mechanism based on the adsorption of MoO_4^{2-} ions is more likely to be valid, because the linear relation, represented by Equation (3), between the logarithmic concentration of Cl^- and the logarithmic concentration of MoO_4^{2-} which is required to inhibit pitting, suggests the adsorption of MoO_4^{2-} ions according to the Freundlich-type adsorption isotherm. Judging from the coefficient (1.17) of the logarithmic term in Equation (3), the inhibiting efficiency of MoO_4^{2-} ions to the chloride pitting of austenitic stainless steels is thought to be lower than that of NO_3^- ions but higher than that of acetate ions.²⁰

Interaction Between Molybdenum in Steel and MoO_4^{2-} Ions in Solution

The common features observed in the pitting inhibition by alloying Mo in steels and adding MoO_4^{2-} ions to chloride solutions are summarized as follows:

1. The pitting potential is shifted in the noble direction with increase in Mo content of steels or in MoO_4^{2-} concentration in solutions.
2. It is impossible to inhibit the initiation of pitting by either method at elevated temperatures. The critical temperature for pitting initiation becomes higher with increasing Mo content in steels or MoO_4^{2-} concentration in solutions.

3. Pitting of the Mo-containing steels can be inhibited by small additions of MoO_4^{2-} ions to solutions by which pitting of the Mo-free steel cannot be suppressed.

Consequently, it is reasonable to assume the existence of a common process in mechanisms of pitting inhibition by the above two methods. Since the formation of MoO_4^{2-} ions, as previously mentioned, can be expected in the initial stage of pitting of the Mo-containing steel, it is presumed that the adsorption of MoO_4^{2-} ions on the active sites within pits plays an important role in the inhibition of pitting. That is, the adsorption process of MoO_4^{2-} ions, which suppress the pit growth, is thought to be common in both methods.

Pitting of Mo-containing Steels in NaBr Solutions

Although the rates of pitting dissolution of the 20Cr-25Ni steel without Mo in NaBr solutions is decreased by the addition of MoO_4^{2-} ions, the inhibiting efficiency of the ions to bromide pitting is extremely low and an almost tenfold addition of MoO_4^{2-} ions is required for complete inhibition of bromide pitting compared with the case for the chloride pitting. The amount of MoO_4^{2-} ions which are concentrated within a pit at the initial stage of pitting of the Mo-containing steel is thought to be almost equal in both cases for chloride and bromide solutions. Consequently, even the 20Cr-25Ni-5Mo steel, which shows higher resistance to pitting in NaCl solutions, seems to be liable to pitting in NaBr solutions.

Oxidation States of Constituents of Passive Film Formed in Chloride Solutions

Oxidation states of the film constituents of specimens A, B, C, and D in Table 2 were estimated from the values of binding energy at the peaks of the photoelectron spectra. The oxidation states of Fe, Cr, Ni and O in surface films were determined on the basis of our previous data in which the photoelectron spectra for the passivation 18Cr-8Ni steel and various kinds of pure oxides, hydroxides, and oxihydroxides were studied in detail.¹⁵ The oxidation states of Mo and Cl were determined from the photoelectron spectra obtained from pure Mo, molybdenum compounds, and sodium chloride in the present study. Literature values for the photoelectron spectra of several molybdenum compounds were also used for comparison.²¹

The characteristic features of photoelectron spectra for specimens A, B, C, and D, and the oxidation states of their passive film constituents are summarized as follows:

Fe. A strong peak at about 712 eV which is usually observed on the Fe 2p_{3/2} spectra of these specimens can be assigned to oxidized Fe [approximately Fe(III)] in the passive films.

Cr. A strong peak at about 577 eV for the Cr 2p_{3/2} spectra of these specimens can be assigned to oxidized Cr [approximately Cr(III)] in the passive films.

Ni. Counting rates of Ni 2p_{3/2} spectra for the stainless steel specimens were very low, suggesting that the Ni content of their passive films is extremely low. The strong peak at about 856 eV is thought to result from oxidized Ni [approximately Ni(II)] in the passive films.¹⁵ A weak peak at about 862 eV can be assigned to the peak due to "shake-up" in the Ni(II) oxide.

O. O 1s spectra of passive films always have two peaks at about 532 and 531 eV, which are attributed to oxygen in the M-OH (M: metal element) or O-OH bond and that in the M-O bond, respectively.¹⁵ Since the intensity of the 531 eV peak is stronger than that of the 532 eV peak in O 1s spectra of specimens A, B, and C, the passive film formed by the anodic oxidation should be composed of an oxide which contains a small amount of hydroxy groups.

Cl. The binding energy of Cl 2p_{3/2} electrons for specimen A, 198.2 eV, is almost equal to that for NaCl powder, 198.5 eV. However, the values of binding energy for Cl 2p_{3/2} electrons of specimens B and C, which are around 200.2 eV, are approximately 2 eV higher than those for NaCl powder, suggesting that weakly adsorbed Cl_2 or Cl exists in the specimen surface. Such Cl_2 or Cl may be produced by prolonged exposure of the specimens to X-ray radiation.²²

Mo. Mo $3d_{3/2}-3d_{5/2}$ spectra obtained from specimens B and D show four peaks at about 235.5, 232.2, 230.9, and 227.8 eV. Among these peaks, the 230.9 and the 227.8 eV peaks are thought to result from metallic Mo or extremely lower valent Mo, because they correspond to the peaks which appeared at 231.0 and 227.9 eV in the Mo $3d_{3/2}-3d_{5/2}$ spectra of pure Mo which were measured immediately after polishing with emery paper. The other two peaks at 235.5 and 232.2 eV correspond to the peaks observed at about 235.8 and 232.7 eV in Mo $3d_{3/2}-3d_{5/2}$ spectrum obtained from powders of MoO_3 and $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$. Consequently, Mo in the passive film of the 20Cr-25Ni-5Mo steel should be oxidized to approximately Mo(VI) when subjected to anodic oxidation or air-oxidation.

A very clear spectrum of Mo $3d_{3/2}-3d_{5/2}$ photoelectrons was obtained from specimen C, i.e., the 20Cr-25Ni steel treated in the $\text{NaCl} + \text{Na}_2\text{MoO}_4$ solution. Since specimen C contains no Mo in the matrix, the spectrum of Mo obtained is thought to result from MoO_4^{2-} ions adsorbed or included in the passive film from the solution used. Two peaks are observed at 235.6 and 232.3 eV on the spectrum, and these two peaks agree with the 235.6 and the 232.2 eV peaks in the Mo $3d_{3/2}-3d_{5/2}$ spectrum for $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$ powder. It is, therefore, likely that the passive film which is formed by anodic oxidation in a NaCl solution with MoO_4^{2-} ions contains Mo(VI).

Finally, it is concluded that Mo in passive films exists as Mo(VI) ions in both cases of the pitting inhibition, that is, the alloying addition of Mo in steels and the addition of MoO_4^{2-} ions to chloride solutions.

Summary

It was found that MoO_4^{2-} ions added to a neutral NaCl solution acts effectively as a pitting inhibitor for the 20Cr-25Ni steel without Mo. Both pit initiation and growth are suppressed by small additions of MoO_4^{2-} ions. Several similar effects are observed in both methods for pitting inhibition, that is, alloying Mo in steels and adding MoO_4^{2-} ions to chloride solutions. Moreover, the multiplied effect in the inhibition of pitting between Mo in the steel and MoO_4^{2-} ions in the solution is recognized.

From results of the surface analysis by X-ray photoelectron spectroscopy (ESCA), it is found that Mo(VI) ions in the passive film participate in the inhibition of the chloride pitting of austenitic stainless steel by alloyed Mo or added MoO_4^{2-} ions, or both.

On the basis of these facts, it is reasonable to assume that the resistance against pitting of Mo-containing austenitic stainless steels in neutral halide solutions results from the inhibition of pit growth by the adsorption of MoO_4^{2-} ions which is concentrated in pits at the initial stage of pitting.

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